

Homework # 4: Utility theory and applications. Standard loss functions and corresponding decisions

Due February 21

1. (Berger, 2.3) Determine and sketch your own personal utility function for money over the interval from \$-1000 and \$1000. Show the main steps in your construction.
2. (Berger, 2.4) Look at the following three pairs of gambles:

$$\begin{array}{lll} \text{The 1st pair:} & \frac{1}{4}\langle 250 \rangle + \frac{3}{4}\langle 0 \rangle & \text{or } \frac{1}{2}\langle 40 \rangle + \frac{1}{2}\langle 70 \rangle \\ \text{The 2nd pair:} & \frac{1}{2}\langle 400 \rangle + \frac{1}{2}\langle -100 \rangle & \text{or } \frac{2}{3}\langle 150 \rangle + \frac{1}{3}\langle 0 \rangle \\ \text{The 3rd pair:} & \frac{1}{2}\langle -1000 \rangle + \frac{1}{2}\langle 1000 \rangle & \text{or } \frac{1}{2}\langle -50 \rangle + \frac{1}{2}\langle 50 \rangle \end{array}$$

- (a) Explain the meaning of gambles in the first pair. How much can you gain with this gamble, and what are the chances?
 - (b) Without referring to your utility function from the previous exercise, decide which gamble you prefer in each of the following pairs.
 - (c) Now find which gamble would be preferred, as determined by your utility function constructed in the previous exercise. Is there any discrepancy between your answers in (b) and (c)? If yes, then you may revise your sketch, but for this homework, it is not required.
3. (Berger, 2.7bd) A person is given a stake of $m > 0$ dollars, which he can allocate between an event A of fixed probability $\alpha \in (0, 1)$ and its complement $\bar{A}$. Let $x \in [0, m]$ be the amount he allocates to A . Then he allocates $(m - x)$ dollars to $\bar{A}$.

The person's reward is the amount he has allocated to either A or $\bar{A}$, whichever actually occurs. Thus he can choose among all gambles of the form

$$\alpha\langle x \rangle + (1 - \alpha)\langle m - x \rangle.$$

Being careful to consider every possible pair of values of α and m , find the optimal allocation of the m dollars when the person's utility function U is defined on the interval $[0, m]$ as follows:

- (a) $U(r) = r$
 - (b)* $U(r) = \log(r + 1)$ (*This question is optional, not required... it requires some Calculus*)
4. (Berger, 2.10) An investor has \$1000 to invest in speculative stocks. He is considering investing m dollars in stock A and $(1000 - m)$ dollars in stock B. An investment in stock A has a 0.6 chance of doubling in value, and a 0.4 chance of being lost. An investment in stock B has a 0.7 chance of doubling in value, and a 0.3 chance of being lost. The investor's utility function is

$$U(x) = \log(0.0007x + 1) \text{ for } -1000 \leq x \leq 1000.$$

- (a) For fixed m , list all the possible rewards and the corresponding probabilities.
 - (b)* Find the optimal value of m in terms of the expected utility. (*Do it only if you did #3b*)
- (*This perhaps indicates why most investors opt for a diversified portfolio of stocks.*)

5. Let $\mathbf{X}$ be a sample of size n from $\text{Uniform}(0, \theta)$ distribution, where θ has $\text{Pareto}(\alpha, \sigma)$ prior.
 - (a) Derive the Bayes estimator of θ under the squared-error loss function.
 - (b) Derive the Bayes estimator of θ under the absolute-error loss function.
 - (c) Derive the generalized maximum likelihood estimator of θ .
6. Let $\mathbf{X}$ be a sample of size n from $\text{Normal}(\theta, \sigma)$ distribution, where σ is known and θ has a non-informative improper prior $\pi(\theta) \equiv 1$.
 - (a) Derive the Bayes estimator of θ under the squared-error loss function.
 - (b) Derive the Bayes estimator of θ under the absolute-error loss function.
 - (c) Derive the generalized maximum likelihood estimator of θ .
7. This is a multi-dimensional problem. Let $\mathbf{X}$ be a $\text{Multinomial}(n; \theta_1, \dots, \theta_k)$ random vector, where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_k)$ has a Dirichlet prior distribution $\pi(\theta_1, \dots, \theta_k \mid \alpha_1, \dots, \alpha_k)$.
 - (a) Derive the (multivariate) posterior mean. It is the Bayes estimator of $\boldsymbol{\theta}$ under the squared-error loss $L(\boldsymbol{\theta}, \boldsymbol{\delta}) = \|\boldsymbol{\theta} - \boldsymbol{\delta}\|^2 = \sum_i (\theta_i - \delta_i)^2$.
 - (b)* Derive the generalized maximum likelihood estimator of $(\theta_1, \dots, \theta_k)$.
(This question is optional... It needs constrained optimization by the method of Lagrange multipliers... Calculus I material)

Multinomial random vector, a generalization of a Binomial random variable, is a vector of counts of n independent events falling into k categories, with θ_i being the probability for an event to fall into category i . For example, if students get an A, a B, a C, and an F in some course with probabilities 0.3, 0.4, 0.2, and 0.1, respectively, then in a group of 20 students, the number of students with each grade is $\text{Multinomial}(20; 0.3, 0.4, 0.2, 0.1)$. Multinomial vector has the probability mass function

$$f(\mathbf{x} \mid \theta_1, \dots, \theta_k) = \frac{n!}{\prod_{i=1}^k x_i!} \prod_{i=1}^k \theta_i^{x_i}, \quad 0 \leq x_i \leq n, \quad \sum_{i=1}^k x_i = n,$$

with $\mathbf{E}(X_i) = n\theta_i$, $\text{Var}(X_i) = n\theta_i(1 - \theta_i)$, $\text{Cov}(X_i, X_j) = -n\theta_i\theta_j$.

Dirichlet distribution, a multidimensional analogue of Beta, has density

$$\pi(\theta_1, \dots, \theta_k \mid \alpha_1, \dots, \alpha_k) = \frac{\Gamma(\alpha_0)}{\prod_{i=1}^k \Gamma(\alpha_i)} \prod_{i=1}^k \theta_i^{\alpha_i-1}, \quad 0 \leq \theta_i \leq 1, \quad \sum_{i=1}^k \theta_i = 1, \quad \alpha_0 = \sum_{i=1}^k \alpha_i,$$

with $\mathbf{E}(\theta_i) = \alpha_i/\alpha_0$ and $\text{Var}(\theta_i) = (\alpha_0 - \alpha_i)\alpha_i/\alpha_0^2(\alpha_0 + 1)$.